

## UNSW ZZSC9020 - Data Science Project - Task Planner (Group A)

| Task Board |                                                                                              |                                |                                  |             |   |         |          |                                                                                                                                                                                                                                                                                                                                        |              |              |                   |  |
|------------|----------------------------------------------------------------------------------------------|--------------------------------|----------------------------------|-------------|---|---------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|--------------|-------------------|--|
|            | Tasks                                                                                        | Phase                          | Assignee                         | Status      | # | LOE (D) | Priority | Description                                                                                                                                                                                                                                                                                                                            | Start Date   | End Date     | Assessment        |  |
| 1          | Team Contract, roles, tooling setup and finalise Research Topic                              | Project Planning               | All Members                      | Complete    | 1 |         | High     | Agree the working contract and decision rules, confirm roles, and stand up Teams, Planner, OneDrive and the GitHub repository.                                                                                                                                                                                                         | Sep 1, 2026  | Sep 3, 2026  | A1A - Plan        |  |
| 2          | Weekly Lecturer Consultation                                                                 | Project Planning               | All Members                      | In Progress | 7 |         | Medium   | Book the slot, circulate an agenda to the repository 24 hours before, publish minutes within 24 hours after.                                                                                                                                                                                                                           | Sep 3, 2026  | Oct 16, 2026 | All               |  |
|            | Project Plan S1: Introduction and Motivation                                                 | Project Planning               | Daniel Cao                       | Complete    | 2 |         | Medium   | Write section 1: the business problem, why medium-term demand forecasting matters, who the client is, and what the project sets out to achieve.                                                                                                                                                                                        | Sep 1, 2026  | Sep 6, 2026  | A1A - Plan        |  |
| 4          | Project Plan S2: Brief Literature Review                                                     | Project Planning               | Arman Hajisafi<br>Kelly Hamilton | Complete    | 4 |         | High     | Write section 2: survey existing work on medium- and long-term load forecasting, identify the methodological gap, and state how this project addresses it.                                                                                                                                                                             | Sep 1, 2026  | Sep 6, 2026  | A1A - Plan        |  |
| 5          | Project Plan S3: Methods, Software and Data Description                                      | Project Planning               | Faiza Mumtaz<br>Swapnil Shinde   | Complete    | 4 |         | High     | Write section 3: the intended models & why they suit the question, software stack, and a description of the three datasets including format, size and known quality issues.                                                                                                                                                            | Sep 1, 2026  | Sep 6, 2026  | A1A - Plan        |  |
| 6          | Project Plan S4: Activities and Schedule                                                     | Project Planning               | Xavier Sun                       | Complete    | 2 |         | Medium   | Write section 4: the activity list, the phase structure, role justification, the timeline and the Gantt chart.                                                                                                                                                                                                                         | Sep 1, 2026  | Sep 6, 2026  | A1A - Plan        |  |
| 7          | Group Review Draft - Project Plan                                                            | Project Planning               | All Members                      | In Progress | 1 |         | Medium   | Read the four sections together, resolve overlaps and contradictions, check the three-page limit, and submit.                                                                                                                                                                                                                          | Sep 6, 2026  | Sep 6, 2026  | A1A - Plan        |  |
| 8          | Complete Individual Implementation Checklist                                                 | Project Planning               | All Members                      | In Progress | 1 |         | High     | Assessment 1B: Each member's checklist by 5 pm Tuesday of Week 2.                                                                                                                                                                                                                                                                      | Sep 5, 2026  | Sep 8, 2026  | A1B - Checklist   |  |
| 9          | Complete Individual Weekly Self-Reflection                                                   | Reporting & Communication      | All Members                      | In Progress | 4 |         | Medium   | An individual task: write a 500-word summary reflective portfolio, based on your weekly teamwork reflections for Weeks 1-4.                                                                                                                                                                                                            | Aug 31, 2026 | Sep 29, 2026 | A2 - Reflection   |  |
| 10         | Ingest validate and clean datasets                                                           | Data Engineering & Preparation | Daniel Cao<br>Kelly Hamilton     | Not Started | 2 |         | Medium   | Load the three CSVs and confirm columns, types, units, date coverage and row counts against the course brief, check duplicates, and resample temperature onto the 30-minute demand grid.<br>External Data: @Kelly to source Solar-Radiation dataset and dataset clean-up.                                                              | Sep 7, 2026  | Sep 9, 2026  | A3 - Report       |  |
| 11         | Build feature set and design rolling-origin splits                                           | Data Engineering & Preparation | Arman Hajisafi                   | Not Started | 3 |         | Medium   | Calendar and public + school-holiday flags, Fourier seasonality terms, a long-run trend and cooling/heating degree days on a fixed 18 C base, which activity 13 then validates rather than supplies; set the train and test origins for each horizon, then review every feature for information that would not exist at forecast time. | Sep 9, 2026  | Sep 12, 2026 | A3 - Report       |  |
| 12         | Load profiles and seasonality                                                                | Exploratory Data Analysis      | Faiza Mumtaz                     | Not Started | 2 |         | Medium   | Demand by hour, weekday, month and year; STL decomposition and autocorrelation to size the seasonal structure the models have to capture.                                                                                                                                                                                              | Sep 9, 2026  | Sep 11, 2026 | A3 - Report       |  |
| 13         | Temperature-demand response and trend and structural breaks                                  | Exploratory Data Analysis      | Swapnil Shinde                   | Not Started | 3 |         | Medium   | Fit the U-shaped response curve, estimate the heating and cooling thresholds, and quantify how many megawatts each additional degree costs. Quantify the daytime decline driven by rooftop solar across 2010-2021 (isolate the 2020 COVID-19 period).                                                                                  | Sep 9, 2026  | Sep 12, 2026 | A3 - Report       |  |
| 14         | Baseline Modelling                                                                           | Modelling                      | Xavier Sun                       | Not Started | 3 |         | High     | Seasonal naive (same period last year) and a climatology profile averaged over prior years. These set the bar every other model has to clear, and every later result is reported relative to them.                                                                                                                                     | Sep 12, 2026 | Sep 15, 2026 | A3 - Report       |  |
| 15         | Statistical models - SARIMAX                                                                 | Modelling                      | Swapnil Shinde                   | Not Started | 3 |         | High     | Fit SARIMAX with temperature as an exogenous input, with ETS and Prophet as comparators; check residual diagnostics and report the fitted seasonal structure.                                                                                                                                                                          | Sep 12, 2026 | Sep 15, 2026 | A3 - Report       |  |
| 16         | ML Modelling - Random Forest                                                                 | Modelling                      | Daniel Cao                       | Not Started | 3 |         | High     | Train a random forest on the horizon-safe feature set, tuned only on validation origins so the test origins stay untouched; record feature importances for the discussion.                                                                                                                                                             | Sep 12, 2026 | Sep 15, 2026 | A3 - Report       |  |
| 17         | ML Modelling - XGBoost                                                                       | Modelling                      | Faiza Mumtaz                     | Not Started | 3 |         | High     | Train a gradient-boosted tree ensemble with XGBoost on the same features and splits; tune depth, learning rate and regularisation on validation origins only.                                                                                                                                                                          | Sep 12, 2026 | Sep 15, 2026 | A3 - Report       |  |
| 18         | ML Modelling - LightGBM                                                                      | Modelling                      | Arman Hajisafi                   | Not Started | 3 |         | High     | Train a LightGBM ensemble on the same features and splits, and compare its training cost and accuracy against XGBoost, which uses the same algorithm family with a different split-finding strategy.                                                                                                                                   | Sep 12, 2026 | Sep 15, 2026 | A3 - Report       |  |
| 19         | ML Modelling - LSTM                                                                          | Modelling                      | Kelly Hamilton                   | Not Started | 3 |         | High     | Train an LSTM sequence model on the same features and splits, and report whether the extra capacity earns its cost at these horizons.                                                                                                                                                                                                  | Sep 12, 2026 | Sep 15, 2026 | A3 - Report       |  |
| 20         | Model evaluation & validation                                                                | Evaluation & Validation        | All Members                      | Not Started | 2 |         | Medium   | Each member puts their own model through the shared harness: fit at every rolling origin, predict the horizon, collect MAE, RMSE and MAPE. Then break the error down by season, temperature band and demand level, test summer peaks and extreme-heat days.                                                                            | Sep 16, 2026 | Sep 18, 2026 | A3 - Report       |  |
| 21         | Model selection and freeze                                                                   | Evaluation & Validation        | All Members                      | Not Started | 1 |         | Medium   | Compare the six models on the common metrics, choose the best per horizon, record the decision and the evidence behind it, and stop experimenting.                                                                                                                                                                                     | Sep 19, 2026 | Sep 20, 2026 | A3 - Report       |  |
| 22         | Selected model improvement                                                                   | Evaluation & Validation        | Model Winner                     | Not Started | 2 |         | Medium   | The member whose model wins takes one focused round of tuning or feature refinement, with the rest of the team reviewing. The owner is decided at Task 21 (Model Selection).                                                                                                                                                           | Sep 21, 2026 | Sep 23, 2026 |                   |  |
| 23         | Report skeleton and section owners, and draft the report chapters                            | Reporting & Communication      | All Members                      | Not Started | 1 |         | Medium   | Agree headings, page budget and a named owner for each of the seven report chapters, then start drafting against the frozen results.                                                                                                                                                                                                   | Sep 21, 2026 | Sep 22, 2026 | A3 - Report       |  |
| 24         | Group Report: Introduction + Conclusion and Further Issues + Organising Reference & Appendix | Reporting & Communication      | Arman Hajisafi                   | Not Started | 5 |         | Medium   | <ul style="list-style-type: none"><li>Chapter 1: Introduction</li><li>Chapter 7: Conclusion and Further Issues</li><li>Reference + Appendix: Sorting &amp; Organising</li></ul>                                                                                                                                                        | Sep 23, 2026 | Oct 7, 2026  | A3 - Report       |  |
| 25         | Group Report: Literature Review + Discussion                                                 | Reporting & Communication      | Kelly Hamilton                   | Not Started | 5 |         | High     | <ul style="list-style-type: none"><li>Chapter 2: Literature Review</li><li>Chapter 6: Discussion</li></ul>                                                                                                                                                                                                                             | Sep 23, 2026 | Sep 29, 2026 | A3 - Report       |  |
| 26         | Group Report: Material and Methods: 3.1, 3.2 and 3.3                                         | Reporting & Communication      | Swapnil Shinde                   | Not Started | 5 |         | Medium   | <ul style="list-style-type: none"><li>Chapter 3:</li><li>3.1 Software</li><li>3.2 Description of the Data</li><li>3.3 Pre-processing Steps</li></ul>                                                                                                                                                                                   | Sep 23, 2026 | Sep 29, 2026 | A3 - Report       |  |
| 27         | Group Report: Material and Methods: 3.4, 3.5 and 3.6                                         | Reporting & Communication      | Daniel Cao                       | Not Started | 5 |         | High     | <ul style="list-style-type: none"><li>Chapter 3:</li><li>3.4 Data Cleaning</li><li>3.5 Assumptions</li><li>3.6 Modelling Methods</li></ul>                                                                                                                                                                                             | Sep 23, 2026 | Sep 29, 2026 | A3 - Report       |  |
| 28         | Group Report: Exploratory Data Analysis                                                      | Reporting & Communication      | Faiza Mumtaz                     | Not Started | 5 |         | Medium   | Chapter 4, written from the EDA notebooks, with every figure regenerated from the cleaned data rather than pasted.                                                                                                                                                                                                                     | Sep 23, 2026 | Sep 29, 2026 | A3 - Report       |  |
| 29         | Group Report: Analysis and Results                                                           | Reporting & Communication      | Xavier Sun                       | Not Started | 5 |         | High     | Chapter 5: the model comparison, the stratified error tables, and what they say about which model wins under which conditions.                                                                                                                                                                                                         | Sep 26, 2026 | Oct 2, 2026  | A3 - Report       |  |
| 30         | Internal review and reproducibility check, and final submission                              | Reporting & Communication      | All Members                      | Not Started | 4 |         | Medium   | Clone the repository fresh, re-run the pipeline end to end, confirm every figure and number in the report regenerates, then submit.                                                                                                                                                                                                    | Sep 29, 2026 | Oct 7, 2026  | A3 - Report       |  |
| 31         | Video storyboard and slides                                                                  | Reporting & Communication      | All Members                      | Not Started | 2 |         | Low      | Storyboard fifteen minutes for a non-technical audience and build one shared deck rather than six.                                                                                                                                                                                                                                     | Oct 7, 2026  | Oct 10, 2026 | A4 - Presentation |  |
| 32         | Record, edit and submit the video                                                            | Reporting & Communication      | Xavier Sun                       | Not Started | 2 |         | Medium   | Each member records their own section; edit the pieces together and submit by 10 pm Tuesday of Week 7.                                                                                                                                                                                                                                 | Oct 10, 2026 | Oct 13, 2026 | A4 - Presentation |  |
| 33         | Project coordination                                                                         | Project Planning               | Xavier Sun                       | In Progress | 4 |         | Medium   | Maintain the task board and Gantt, keep the risk log current, write the weekly agendas and minutes, and enforce the branch-and-review convention on the repository.                                                                                                                                                                    | Aug 31, 2026 | Oct 13, 2026 | All               |  |

+



UNSW ZZSC9020 - Data Science Project - Task Planner Timeline (Group A)

